MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000016

(RANGE SERIES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Range Series Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-24308/1	Connectors, Electric, Rectangular, Miniature Polarized Shell, Rack and Panel General Specification for
MIL-C-50785	Circuit Card Assembly, Sight Current Generator, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

12000239	Sight Current Generator
12000016	Circuit Card Assembly, Range Series
12000091	Microcircuit, Multiply/Divide

PACKAGING DATA SHEET

12000016	Packaging of Circuit Card Assembly, Range Series

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(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is a single-ended printed circuit board consisting of three transistors, six operational amplifiers and three divider modules. The assembly is used to generate the range series input term of the magnet current voltage circuitry when installed in the Sight Current Generator Drawing Number 12000239.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000016.

3.1.2 Government furnished property list.- Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing requirements a through d when assembled in accordance with Drawing Number 12000016.

a. Operational amplifiers AR1 and AR2 in conjunction with Q1 and Q2 shall generate a +7 Vdc reference voltage, which is based on the -10 Vdc radar reference voltage, when the sight current generator is operating in the radar mode and -7 Vdc when the sight current generator is operating in the manual mode.

b. Operational amplifiers AR3 and AR4 in conjunction with Q3 shall amplify and scale the range input voltage.

c. Operational amplifier AR6 shall generate the +7 Vdc reference voltage.

d. Dividers U1, U2 and U3 in conjunction with operational amplifier AR5 shall develop and scale the range correct term.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- Input power shall be supplied to the assembly as specified in Table I.

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TABLE I - INPUT VOLTAGES

PIN	VOLTAGE
14 and 15	+15 Vdc ± 2.0 percent
3 and 4	-15 Vdc ± 2.0 percent
1 and 2	Ground
21	-7 Vdc ± 0.1 percent
5 or 6	Triangle wave -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 1.00 milliseconds ± 10 percent.
7 or 8	Triangle wave -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of .8 millisecond ± 10 percent.

3.2.1.1.2 Output levels.- The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.10 shall be met when supplied with the specified voltage input. The inputs applied to the microcircuits U1, U2 and U3 shall not exceed the electrical requirements specified on Drawing Number 12000091.

3.2.1.1.2.1 Pin 9 output.- The output between pin 9 and pin one shall be 1.568 (U1) - 1.425 (U2) + 0.730 (U3) ± 4.0 percent with a Vin applied between pin 11 and pin one where:

$$U1 = \frac{4.9}{(-)\ (0.5833)(Vin)}$$

$$U2 = \frac{0.7U1}{(-)\ (0.5833)(Vin)}$$

$$U3 = \frac{0.7U2}{(-)\ (0.5833)(Vin)}$$

Vin = + 1.0 Vdc to + 12 Vdc with + 15 Vdc applied to pin 18
or, Vin = + 1.25 Vdc to + 15 Vdc with - 15 Vdc applied to pin 18

3.2.1.1.2.2 Pin 13 output.- The output between pin 13 and pin one shall be as specified in a, b and c.

a. Zero ± .006 Vdc with zero Vdc applied between pin 11 and pin one, and -15 Vdc applied between pin 18 and pin one.

b. Vin (+0.466) ± 1.0 percent with Vin of zero to +15 Vdc applied between pin 11 and pin one, and -15 Vdc applied between pin 18 and pin one.

c. Vin (+0.583) ± 1.0 percent with Vin of zero to +15 Vdc applied between pin 11 and pin one, and +15 Vdc applied between pin 18 and pin one.

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3.2.1.1.2.3 Pin 22 output.- The output between pin 22 and pin one shall be as specified in a and b.

a. +7.0 ± 0.020 Vdc with -10 Vdc applied between pin 16 and pin one, -7.0 Vdc applied between pin 21 and pin one and + 13 Vdc applied between pin 18 and pin one.

b. +7.0 ± 0.020 Vdc with -7.0 Vdc applied between pin 21 and pin one, and -15 Vdc applied between pin 18 and pin one.

3.2.1.1.2.8 Pin 23 output.- The output between pin 23 and pin one shall be +7.0 Vdc ± 0.1 percent when -7.0 Vdc ± .01 percent is applied between pin 21 and pin one.

3.2.1.2 Fault Isolation Test Points.- The assembly has the following test points available to facilitate fault isolation. Pins 10, 12, 17, 20, 24, 25, TP1-6. These are not output pins.

3.2.1.3 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristics.- The physical characteristics shall be as specified in drawing 12000016 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000016 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

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4.2.1.1 Power supply (PS1).- The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS3).- The power supply shall be zero to +15 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Power supply (PS4).- The power supply shall be -10 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) RMS or 100 mv peak-to-peak.

4.2.1.5 Power supply (PS5).- The power supply shall be -7 ± 0.001 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) or 100 mv peak-to-peak.

4.2.1.6 Power supply Ø1 (PS6).- The power supply shall generate a triangle wave -8.47 ± 1.03 Vdc to +8.47 ±1.03 Vdc within a period of 1.00 millisecond ± 10 percent.

4.2.1.7 Power supply Ø2 (PS7).- The power supply shall generate a triangle wave -8.47 ±1.03 Vdc to +8.47 ±1.03 Vdc within a period of 0.8 millisecond ±10 percent.

4.2.1.8 Digital voltmeter (DVM). The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 1 using receptacle M24308/1-3 of specification MIL-C-24308-1.

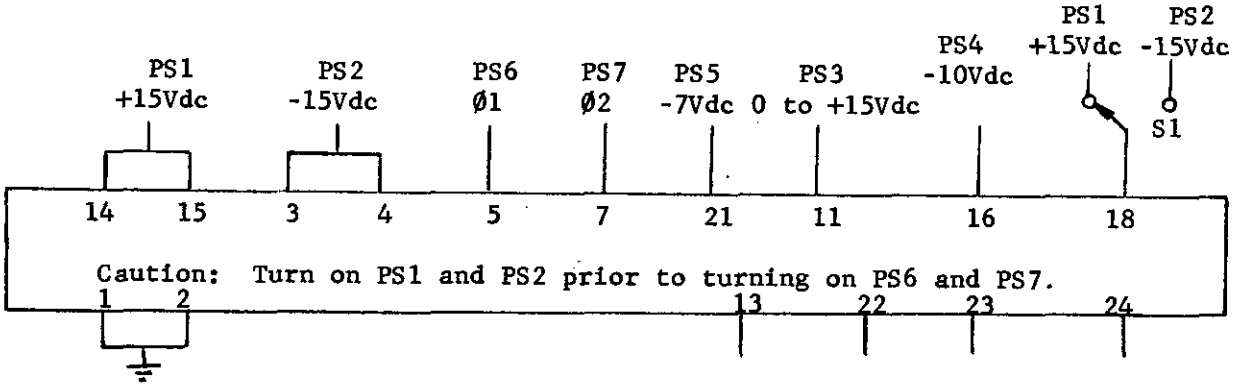


FIGURE 1 - TEST CONNECTIONS

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4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.2 through 3.2.1.1 shall be verified as follows:

a. Set switch S1 to +15 Vdc, adjust power supplies PS3 and PS4 as specified in Table II.

TABLE II - CONNECTIONS

Pin 11 Input Vdc	Pin 16 Input Vdc	Pin 9 Output Vdc
1.2 ±0.007	-10 ±0.007	6.110 ±0.245
4.0 ±0.007	-10 ±0.007	2.533 ±0.100
8.0 ±0.007	-10 ±0.007	1.441 ±0.058
12 ±0.007	-10 ±0.007	0.997 ±0.040

b. Set switch S1 to -15 Vdc, adjust power supplies PS3 and PS4 as specified in Table III.

TABLE III - CONNECTIONS

Pin 11 Input Vdc	Pin 16 Inputs Vdc	Pin 9 Output Vdc
1.5 ±0.007	-10 ±0.007	6.110 ±0.245
15 ±0.007	-10 ±0.007	0.997 ±0.025

c. Set switch S1 to +15 Vdc, adjust power supply PS3 as specified in Table IV.

TABLE IV - CONNECTIONS

Pin 11 Input Vdc	Pin 13 Output Vdc
0 ±0.001	0 ±0.006
12 ±0.007	7 ±0.070

d. Set switch S1 to -15 Vdc, adjust power supply PS3 as specified in Table V.

TABLE V - CONNECTIONS

Pin 11 Input Vdc	Pin 13 Output Vdc
0 ±0.001	0 ±0.006
15 ±0.007	7 ±0.070

e. Set switch S1 to +15 Vdc. The output measured at pin 22 with the DVM shall be +7 ±0.020 Vdc.

f. Set switch S1 to -15 Vdc. The output measured at pin 22 with the DVM shall be +7 ±0.020 Vdc.

g. The voltage measured at pin 23 with the DVM shall be +7 ±0.007 Vdc.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000016.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A435

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: MIL-C-50785/16(MU) Circuit Card Assembly 12000016 (Range Series)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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